STATE OF NEW MEXICO
OFFICE OF THE STATE ENGINEER
Hydrology Bureau

MISCELLANEOUS MEMORANDUM

DATE: June 17, 2025

TO: Ramona Martinez, District VI Manager, Water Rights Division
Lorraine Garcia, Upper Rio Grande Basin Manager, Water Rights Division
Cindy Stokes, Interstate Stream Commission

THROUGH: Katie Zemlick, Chief, Hydrology Bureau, Santa Fe, NM KHS (for Katie Zemlick)

FROM: Brad Wolaver, Senior Hydrologist, Hydrology Bureau, Santa Fe, NM BDW
Laura Petronis, Water Resources Manager 1, Hydrology Bureau, Santa Fe, NM LJP

SUBJECT: Calendar Year 2024 hydrologic impacts on the Rio Grande, Rio Pojoaque-Nambe, Rio Tesuque, and La Cienega springs due to Buckman well field pumping, RG-20516 et al.

KEYWORD: Miscellaneous Memorandum, District VI, Santa Fe County, OSE Rio Grande Basin, City of Santa Fe

ID: MSC_2025_002

SUMMARY AND CONCLUSION

The memorandum presents estimates of the effects of City of Santa Fe (City) Buckman Well Field (Buckman) pumping on the Rio Grande, Rio Pojoaque-Nambe, Rio Tesuque, and La Cienega Springs. Stream depletion to the Rio Grande, Rio Pojoaque-Rio Nambe, and Rio Tesuque is evaluated from 1972 to 2024. Impacts to La Cienega spring discharge are evaluated from 1988 to 2024. All dates are referenced to calendar year pumping (January through December). These depletion estimates support regulatory oversight by the New Mexico Office of the State Engineer (OSE). The OSE Water Rights Division, District 6 evaluates compliance with Buckman well field permit conditions, and the OSE and Interstate Stream Commission (ISC) coordinate to determine whether San Juan-Chama water releases are required to offset Rio Grande surface water depletions.

Key findings of the 2024 evaluation are (see Tables 1–5 for detailed results):

Buckman Pumping in 2024

- Total 2024 pumping: 1372.95 acre-feet (ac-ft), twelfth lowest annual value since 1988

- Wells 10 through 13 accounted for 28.2% of total pumping (386.86 ac-ft)
- The three wells in the model cell located closest to the Rio Grande contributed 52.0% of total pumping (713.75 ac-ft; Buckman 1, 7, and 8; model row 13, column 11)

Stream Depletion in 2024

- Rio Pojoaque-Nambe system: 60.797 ac-ft
- Rio Tesuque: 33.583 ac-ft
- Rio Grande above Otowi gage: 101.43 ac-ft
- Rio Grande below Otowi gage: 842.94 ac-ft

La Cienega Spring Impacts in 2024

- Springs flow reduction: 3.74 ac-ft

SIMULATED STREAM DEPLETIONS ON THE RIO GRANDE AND SANTA FE-AREA TRIBUTARIES

Streamflow depletion on the Rio Grande, Rio Pojoaque-Nambe, and Rio Tesuque due to Buckman pumping from 1972 to 2024 was estimated by following the general procedure described in Compton (1988). It specifies that residual hydrologic effects on the Rio Grande and its tributaries resulting from 1972 through 1987 Buckman pumping be calculated using the NMOSE analytical model (Spinks, 1988), and that post-1987 pumping effects on nearby streams be calculated using the McAda and Wasiolek (1988) ground-water flow model.

Residual hydrologic effects on the Rio Pojoaque-Nambe and Rio Tesuque due to 1972–1987 Buckman pumping were described in Spinks (1988) for 1988 through 2010 using the NMOSE analytical model. Core (2003) projected 1972–1987 Buckman pumping effects on the Rio Tesuque and Rio Pojoaque-Nambe after 2010. Residual hydrologic effects on the Rio Grande due to 1972–1987 Buckman pumping were described in Fleming (1989) using the NMOSE analytical model.

Prior to 2004, the McAda-Wasiolek (1988) model was used to calculate impacts for 1988 through 2003 (based on pumping from 1988 through 2003). For the 2004 calculation (based on pumping from 1988 through 2004), and for subsequent annual calculations, the revised superposition version of the McAda-Wasiolek (1988) model (Core, 1996; Barroll and Keyes, 2005) was used and was run with MODFLOW96. Use of the superposition model has consolidated the number of separate model runs needed, and additionally has allowed for estimation of effects at La Cienega springs (see below). The total diversion for each of the Buckman wells was divided equally between model layers 1 and 2 and monthly pumping was simulated in the model, which is consistent with how pumping has been simulated in previous years. For this analysis, 2024

monthly pumping from the Buckman wells was added to the well file (which contained the previously entered 1988 through 2023 pumping), the model was run, and the results were processed.

Table 3 summarizes the estimated streamflow depletion on the Rio Pojoaque-Nambe and Rio Tesuque from 1988 to 2030 due to Buckman pumping from 1972 to 2024. Stream depletion for 2024 was calculated to be 60.797 ac-ft on the Rio Pojoaque-Nambe system, and 33.583 ac-ft on the Rio Tesuque. Starting in 2024, maximum calendar year depletions based on Buckman pumping from 1972 to 2024 were estimated to be 61.093 ac-ft (estimated to occur in 2025) and 33.583 ac-ft (estimated to have occurred in 2024), in the Rio Pojoaque-Nambe system and the Rio Tesuque, respectively.

Table 4 summarizes the monthly estimated streamflow depletion on the Rio Grande above and below Otowi gage for 2024. For 2024, the total estimated streamflow depletion on the Rio Grande for the reach above Otowi gage is 101.43 ac-ft and for the reach below Otowi gage is 842.94 ac-ft.

SIMULATED FLOW REDUCTION FROM LA CIENEGA SPRINGS

The effect of Buckman pumping on La Cienega spring flow was calculated using the revised superposition version of the McAda-Wasiolek (1988) model (Core, 1996, Barroll and Keyes, 2005). The total diversion for each of the Buckman wells was divided equally between model layers 1 and 2 and monthly pumping was simulated in the model.

Table 5 summarizes the estimated depletion to La Cienega springs from 2004 to 2030 due to Buckman pumping from 1988 to 2024. Buckman pumping from 1988 to 2024 is estimated to have caused a spring flow reduction at La Cienega of 3.74 ac-ft during 2024. Based on pumping during this period, the hydrologic effect on the La Cienega Springs would reach 4.80 ac-ft in 2030 and the maximum impact from historical pumping starting in 1988 reaches 16.81 ac-ft in year 2152.

REFERENCES

Barroll, Peggy and Keyes, Eric (2005) Resolution of Discrepancy between Superposition and Calibrated Versions of the Model", November 29, 2005, Office of the State Engineer Technical Memorandum to Tom Morrison, Hydrology Bureau Chief.

Compton, M.B. (1988) 'Untitled letter to Sangre de Cristo Water Company / V. Phillip Soice from S.E. Reynolds, State Engineer, Re: File No. RG-20516 — Buckman Well Field, dated January 14, 1988, State of New Mexico State Engineer Office'.

Core, Andrew B. (1996) 'Espanola Basin, Santa Fe Region Modified McAda-Wasiolek Model User Manual', November 1996 New Mexico State Engineer Office Technical Division Hydrology Report 96-2.

Core, Andrew B. (2003) Residual Effects of 1972-1987 pumping at Buckman well field on the Rio Tesuque and Pojoaque River, New Mexico Office of the State Engineer Memorandum to Tom Morrison, Hydrology Bureau Chief, dated April 30, 2003.

Fleming, William M. (1989) Untitled letter to Mr. Philip Soice, Manager, Sangre de Cristo Water Company , dated June 13, 1988.

McAda, Douglas P. and Wasiolek, Maryann (1988) Simulation of the Regional Geohydrology of the Tesuque Aquifer System near Santa Fe, New Mexico, U.S. Geological Survey Water-Resources Investigations Report 87-4056.

Spinks, Michael P. (1988) Letter from M. Spinks, NMOSE to Phillip Soice, Division Manager, Sangre de Cristo Water Co. dated June 1, 1988.

TABLES

Table 1. Reported Annual Buckman Well Field pumping (acre-feet)

Note that some values differ slightly from those in Table 2 due to differences in rounding.

Well:	1	2	3/3A	4	5	6	7	8	9	10	11	12	13	Total
1988	547.85	742.16	82.49	465.34	52.44	581.03								2,471.31
1989	563.10	1,271.65	260.73	1,049.03	146.30	1,259.71								4,550.52
1990	333.83	840.18	263.18	1,041.18	142.50	1,202.73								3,823.60
1991	200.79	544.79	177.28	625.51	109.60	801.59	301.83	377.50						3,138.89
1992	194.44	729.85	112.18	497.37	48.19	1,215.33	993.01	961.78						4,752.15
1993	692.78	735.49	137.31	647.47	140.81	1,222.32	1,070.03	962.28						5,608.49
1994	707.94	743.24	125.12	667.62	242.34	1,051.71	638.06	777.44						4,953.47
1995	945.40	824.76	351.55	765.18	384.14	1,009.37	861.37	778.63						5,920.40
1996	935.43	707.33	616.53	631.95	437.94	972.46	592.58	781.45						5,675.67
1997	924.65	815.53	467.03	453.35	139.11	1,030.86	497.11	347.11						4,674.75
1998	680.42	878.84	457.50	399.05	110.57	1,042.80	742.71	923.63						5,235.52
1999	624.75	841.62	415.54	353.42	0.00	993.70	1,159.01	890.50						5,278.54
2000	626.28	720.21	519.30	266.50	0.00	914.97	1,169.10	864.24						5,080.60
2001	716.03	489.80	436.06	354.01	0.00	1,037.41	1,032.51	677.52						4,743.34
2002	730.45	583.98	416.69	554.62	265.43	1,272.77	1,145.18	866.53						5,835.65
2003	508.17	540.25	337.84	368.17	234.42	1,187.87	987.07	780.44	292.08	0.03	251.23	259.45	60.07	5,807.09
2004	26.51	6.17	73.44	555.22	3.76	958.14	277.25	652.62	5.67	453.64	710.82	948.33	1,265.27	5,936.84
2005	4.99	6.39	105.14	510.34	1.04	367.40	1,223.04	797.05	1.73	22.76	367.91	216.31	160.25	3,784.35
2006	20.00	150.23	247.19	612.03	7.66	1,076.65	237.57	814.32	54.37	656.51	824.75	88.36	407.43	5,197.07
2007	398.50	8.63	183.29	565.27	0.30	408.79	703.51	704.39	25.12	222.53	136.92	31.49	404.63	3,793.37
2008	75.49	0.00	115.27	782.12	60.02	37.57	851.48	866.44	0.00	262.74	556.46	42.16	84.28	3,734.03
2009	29.56	0.00	143.06	704.14	157.32	951.76	530.90	367.97	0.00	233.39	330.98	345.36	7.89	3,802.33
2010	81.60	0.00	437.16	474.20	128.67	1,023.46	249.29	7.70	0.00	96.65	328.05	181.80	44.33	3,052.91
2011	77.93	0.00	101.38	219.65	0.62	432.63	0.00	244.35	0.00	194.11	288.53	165.30	0.01	1,724.51
2012	213.82	0.00	95.35	149.52	48.83	316.83	0.00	75.33	0.00	20.09	0.00	130.64	0.00	1,050.41
2013	631.14	4.60	218.81	403.80	0.00	514.24	31.48	255.14	4.85	184.96	288.29	159.11	194.03	2,890.45
2014	54.03	0.00	81.58	33.97	29.46	134.59	53.67	20.63	70.81	0.00	29.67	40.50	25.89	574.80
2015	163.55	0.00	56.61	11.98	14.31	163.26	32.89	49.61	7.74	0.01	25.21	29.26	73.34	627.77
2016	231.96	0.00	37.16	85.96	0.01	220.13	89.25	150.83	19.23	28.69	4.41	29.59	27.60	924.82
2017	335.63	0.00	0.01	29.75	0.00	224.68	0.20	67.52	0.04	0.05	5.28	0.00	10.93	674.09
2018	361.44	0.01	16.94	44.58	0.00	160.71	0.10	55.57	14.39	33.94	40.59	14.41	88.48	831.16
2019	95.14	0.15	21.36	16.83	0.74	90.59	1.45	64.00	1.03	13.20	0.03	0.00	9.57	314.09
2020	276.18	0.00	6.64	25.38	0.02	111.77	0.09	127.51	1.82	48.51	15.68	3.20	15.80	632.60
2021	241.72	0.00	3.85	35.38	0.00	74.29	12.25	98.06	25.85	32.08	0.00	10.97	124.48	658.93
2022	458.14	0.01	0.40	38.18	0.00	7.92	0.31	196.50	11.81	45.17	89.00	0.18	127.85	975.47
2023	224.06	0.00	50.15	47.36	0.00	31.36	0.37	125.08	5.83	4.65	0.17	0.14	377.31	866.48
2024	601.27	0.01	56.23	63.34	0.00	152.22	1.00	111.46	0.52	151.41	13.62	9.80	212.02	1,372.90
Average, 1988–2024	392.84	329.35	195.33	393.21	78.56	655.56	455.46	465.92	24.68	122.96	195.80	123.02	169.16	3,269.44
Average, 2022–2024	427.82	0.01	35.59	49.63	0.00	63.83	0.56	144.35	6.05	67.08	34.26	3.37	239.06	1,071.62

Table 2. 2024 Annual Buckman Well Field pumping (acre-feet)

Note that some values differ from those in Table 1 and 2024 total (1,372.95 acre-feet) differs slightly from that reported by the City (1,372.91 acre-feet) due to differences in rounding. The numbers in this table were added to the well file for 2024.

Well	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	Total
1	16.89	38.81	10.88	40.70	41.50	68.45	162.49	124.37	45.14	52.05	0.00	0.00	601.28
2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.01
3	0.00	0.07	0.06	0.07	0.38	23.11	0.79	8.54	0.05	6.38	16.73	0.05	56.23
4	0.00	0.01	0.17	0.08	0.09	24.47	20.16	8.25	3.22	6.90	0.00	0.00	63.35
5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6	0.15	0.20	0.97	0.00	19.53	5.61	0.13	72.10	40.44	13.09	0.00	0.00	152.22
7	0.08	0.11	0.07	0.74	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
8	2.10	0.05	0.87	2.85	26.93	11.13	0.08	1.74	2.63	15.59	0.00	47.50	111.47
9	0.07	0.06	0.06	0.05	0.05	0.05	0.08	0.06	0.05	0.00	0.00	0.00	0.53
10	0.10	0.08	0.00	0.00	0.18	0.07	60.27	44.22	24.09	10.08	0.08	12.24	151.41
11	0.00	0.00	0.00	0.00	0.00	0.06	0.06	0.05	7.30	0.00	3.61	2.55	13.63
12	0.00	0.00	0.00	0.00	0.00	0.08	0.00	5.11	3.03	0.20	0.08	1.30	9.80
13	0.00	0.00	0.00	0.00	0.00	34.99	58.00	69.51	15.65	31.54	0.29	2.04	212.02
Total	19.39	39.39	13.08	44.49	88.66	168.02	302.06	333.95	141.60	135.84	20.79	65.68	1,372.95

Table 3. Estimated impacts from reported Buckman Well Field pumping on the Rio Pojoaque-Nambe and Rio Tesuque through 2024 (acre-feet)

- **Residual impacts from 1972–1987 Buckman pumping** were calculated using the NMOSE analytical model (Spinks, 1988), with input data from Spinks (1988) and Core (2003). Estimated residual impacts decline to near nil by 2017 for the Rio Pojoaque–Nambe, and by 2050 for the Rio Tesuque.
- **Impacts from 1988–2024 Buckman pumping** (with no pumping after 2024) were calculated as follows:
  - **For 1988–2003**, using the McAda and Wasiolek (1988) numerical groundwater flow model (Compton, 1988);
  - **For 2004–2024**, using the Revised Superposition Model (Core, 1996; Barroll and Keyes, 2005); and
  - Model cell placements for wells 5 and 8 were based on GPS measurements taken on April 28, 2004.
- ** Indicates values prior to correction of simulated pumping rates in February 2005 and February 2006; corrected value are shown in parentheses for those years.

	Rio Pojoaque-Rio Nambe			Rio Tesuque		
Year	**Residual Impact of 1972–1987 Pumping (Analytical Model)**	**Impact of 1988–2024 Pumping (Superposition Model)**	**Total Impact**	**Residual Impact of 1972–1987 Pumping (Analytical Model)**	**Impact of 1988–2024 Pumping (Superposition Model)**	**Total Impact**
1988	40.432	1.060	**41.492**	21.015	0.145	**21.160**
1989	39.244	3.287	**42.531**	22.333	0.601	**22.934**
1990	37.971	5.480	**43.451**	23.391	1.166	**24.557**
1991	36.557	7.319	**43.876**	24.227	1.629	**25.856**
1992	34.928	9.960	**44.888**	24.868	2.163	**27.031**
1993	33.112	13.118	**46.230**	25.327	2.867	**28.194**
1994	31.185	15.942	**47.127**	25.615	3.497	**29.112**
1995	29.226	18.910	**48.136**	25.747	4.279	**30.026**
1996	27.296	21.873	**49.169**	25.737	4.929	**30.666**
1997	25.439	24.144	**49.583**	25.608	5.517	**31.125**
1998	23.678	26.656	**50.334**	25.378	6.154	**31.532**
1999	22.028	29.379	**51.407**	25.067	6.820	**31.887**
2000	20.491	32.254	**52.745**	24.691	7.412	**32.103**
2001	19.068	34.714	**53.782**	24.265	7.956	**32.221**
2002	17.753	37.610	**55.363**	23.800	8.659	**32.459**
2003	16.543	40.578	**57.121**	23.308	9.506	**32.814**
2004	15.429	41.763	**57.192**	22.797	10.222	**33.019**
2005	14.404	42.778** (42.781)	**57.182** (57.185)**	22.273	11.294** (11.295)	**33.567** (33.568)**
2006	13.462	44.297** (44.306)	**57.759** (57.768)**	21.743	11.967** (11.968)	**33.680* (33.711)**
2007	12.595	46.327	**58.922**	21.212	12.709	**33.921**
2008	11.797	48.426	**60.223**	20.683	13.363	**34.046**
2009	11.061	50.005	**61.066**	20.157	13.875	**34.032**
2010	10.383	51.723	**62.106**	19.639	14.526	**34.165**
2011	6.151	53.077	**59.228**	19.258	15.163	**34.421**
2012	4.693	53.850	**58.543**	18.767	15.633	**34.400**
2013	3.234	54.410	**57.644**	18.276	15.946	**34.222**
2014	1.775	55.726	**57.501**	17.785	16.474	**34.259**
2015	0.316	56.351	**56.667**	17.295	16.890	**34.185**
2016		56.944	**56.944**	16.804	17.256	**34.060**
2017		57.520	**57.520**	16.313	17.588	**33.901**
2018		58.133	**58.133**	15.822	17.974	**33.796**
2019		58.683	**58.683**	15.331	18.385	**33.716**
2020		59.154	**59.154**	14.841	18.831	**33.672**
2021		59.395	**59.395**	14.350	19.180	**33.530**
2022		59.844	**59.844**	13.859	19.631	**33.490**
2023		60.323	**60.323**	13.368	20.123	**33.491**
2024		60.797	**60.797**	12.877	20.706	**33.583**
2025		61.093	**61.093**	12.387	21.188	**33.575**
2026		60.902	**60.902**	11.896	21.603	**33.499**
2027		60.557	**60.557**	11.405	21.939	**33.344**
2028		60.440	**60.440**	10.914	22.352	**33.266**
2029		60.012	**60.012**	10.424	22.654	**33.078**
2030		59.734	**59.734**	9.933	23.019	**32.952**

Table 4. Estimated 2024 monthly effects on the Rio Grande due to Buckman Well Field pumping from 1972 to 2024 (acre-feet)

The Revised Superposition Model (Core, 1996; Barroll and Keyes, 2005) was used to calculate depletions from pumping during 1988–2024, while depletions due to Buckman Well Field pumping from 1972 to 1987 calculated using the Analytical Model are 0 ac-ft in 2024 based on data in Fleming (1989).

	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	Total
Above Otowi	8.63	7.78	8.59	8.30	8.56	8.28	8.57	8.60	8.35	8.66	8.41	8.70	**101.43**
Below Otowi	55.80	52.20	55.22	55.99	63.12	66.92	85.16	92.40	84.44	85.61	72.73	73.36	**842.94**

Table 5. Estimated impacts on La Cienega Springs due to Buckman Well Field pumping through 2024 (acre-feet)

The Revised Superposition Model (Core, 1996; Barroll and Keyes, 2005) was used to calculate model-predicted streamflow depletion on La Cienega Springs due to simulated 1988–2024 Buckman wellfield pumping (no pumping after 2024). Calculations used model cell placements for Buckman Well Field wells 5 and 8 based on 4/28/2004 GPS measurements.

Year	Total
2004	0.45
2005	0.66
2006	0.83
2007	0.99
2008	1.16
2009	1.32
2010	1.49
2011	1.65
2012	1.82
2013	1.97
2014	2.13
2015	2.29
2016	2.45
2017	2.60
2018	2.75
2019	2.90
2020	3.06
2021	3.21
2022	3.37
2023	3.54
2024	3.74
2025	3.92
2026	4.10
2027	4.27
2028	4.46
2029	4.62
2030	4.80